BIIL: An AIoT and Computer Vision Enabled Industrial Intelligence Platform for Smart RMG Manufacturing in Bangladesh

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Under the supervision of

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SUPERVISOR'S CERTIFICATE

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Abstract

Bangladesh's Ready Made Garment (RMG) and textiles sector forms the backbone of the country's manufacturing economy. A great deal of the industry still operates on manual records, periodic inspections and a range of disparate software packages. As a result machine failure, production stoppages, product defects, energy waste and safety issues can go completely unnoticed until they become very expensive. This project is the development of BIIL, an AIoT and Computer Vision platform for smart RMG manufacturing in Bangladesh. BIIL will use industrial sensors, cameras, edge processing, AI and an Opencv based computer vision pipeline to monitor machine health, production, product quality, energy consumption and workplace safety. The data from all of these sources will be validated, curated, stored and presented via dashboards, reports and alerts. BIIL will be able to pass on selected data to existing ERP and MES systems which the factory is already using. The platform is designed to be modular so it could start with a small pilot and then roll out to additional machines, production lines and/or factory sites. The end goal is a fully functional prototype which will provide factory managers, production supervisors, maintenance engineers, quality inspectors and other authorised personnel with a far better view of what is going on in the factory.

Keywords: Industry 4.0, AIoT, Industrial IoT, Artificial Intelligence, Computer Vision, OpenCV, Smart RMG Manufacturing, Predictive Maintenance, Industrial Monitoring.

Chapter 1

Introduction

Manufacturing is a major part of Bangladesh's economy and the garment/textile factories in particular are highly machine dependent with a large number of machines, workers and production stages. The information is dispersed across machine interfaces, paper records and various software systems making it very difficult to get a picture of the overall state of the factory.

IoT, AI and Computer Vision all provide factories with new ways of gathering and examining the data. Our project the Bangladesh Industrial Intelligence Layer (BIIL) is a platform that brings all of these together.

Project scope. BIIL is for the garment, textile and other manufacturing sectors in Bangladesh. It gathers data from industrial machines, environmental sensors, product scanners and cameras. Factory staff with appropriate authorisation can access a central dashboard to view machine performance, production progress, product quality, energy consumption and workplace safety. The system is designed to support both a small pilot and a larger factory with multiple production lines.

1.1 Problem statement

Factories face several problems in their daily work:

i. Unexpected machine breakdowns which stop working.

ii. It's very difficult to monitor the machines and the production lines in real time.

iii. Product defects which may not be picked up until later on.

iv. High energy consumption and massive amounts of energy wasted.

v. Limited traceability of products throughout the manufacturing process.

vi. Workplace safety risks and the lack of timely alerts.

vii. Factory data stored in various systems.

These are the conditions under which a central monitoring system would be required to assist both staff and management in responding to problems as quickly as possible. The proposal is specifically in the context of the Bangladeshi manufacturing sector and would need to offer real time visibility, early warning of failures and defects and be integrated with the existing ERP/MES software. It would need to be affordable for the local factories, be within the project budget (Section 3.2), and be capable of growing from a small pilot to a larger scale installation.Although some of the factories in Bangladesh do use digital tools, effective real time monitoring is still very limited as the machine, production, quality, energy and safety data are all spread across different systems. This has a direct impact on the factory manager, the supervisor, the maintenance engineer and the quality inspector. The study therefore looks at whether an affordable AIoT and Computer Vision platform can improve operational visibility, early problem detection, productivity and workplace safety.

1.2 Aims, objectives & Motivation

The following sections outline the aim, objectives and motivation for BIIL.

1.2.1 Aims

This project is about designing and building BIIL as an affordable industrial platform for garment, textile and other factories in Bangladesh. IoT, AI and Computer Vision will give factory managers a real time view of machine health, production, product quality, energy usage and workplace safety.

1.2.2 Objectives

i. To monitor machine condition and production activity in real time.

ii. To detect abnormal machine behaviour and predict potential failures.

iii. To detect product defects via Computer Vision (Opencv based inspection).

iv. To track products and production activity via barcode and RFID systems.

v. To monitor energy usage and factory environmental conditions.

vi. To improve worker safety via alerts and PPE monitoring.

vii. To provide dashboards, reports and useful recommendations.

viii. To support integration with existing ERP and MES systems.

1.2.3 Motivation

Garments and textiles is a huge part of Bangladesh's economy but a lot of the factories are still using manual records, have their own separate software and are on a reactive maintenance model. An unexpected failure or a defect that has gone undetected can lead to a delay in an order and increased costs. The smart factory solutions that are out there are generally aimed at the large international plants and so there is a need for a more affordable solution that is appropriate for the local factory situation. We therefore propose a platform that can start off on a single pilot line and then be rolled out with the factory. BIIL is designed to assist the manager in using the data that is already available, to reduce waste and downtime, to protect the workforce and to improve the factory as a whole.

Chapter 2

Literature Review and Analysis

Factories all over the world are using the IoT, AI and Computer Vision to gather information, monitor equipment, reduce production problems and check product quality.

A number of companies have already introduced similar systems in their own plants. These examples form the basis of the proposed BIIL system.

2.1 Existing Applications in Smart Factories

BMW – Predictive Maintenance. BMW have described how factory sensor data can be used to detect equipment problems before a machine actually stops. A change in the power consumption of a conveyor for instance may be an indicator of mechanical wear [1]. BIIL use the same general principle, taking temperature, vibration, current and RPM data to detect anomalies and to support predictive maintenance.

BMW – AI-Based Quality Inspection. BMW uses AI and computer vision in its factories. Its AIQX quality platform looks at images and sensor data from the production lines to detect missing parts, assembly errors etc. [2]

Siemens – Smart Manufacturing in Germany. Siemens is using digital manufacturing technologies at its Electronics Works in Amberg, Germany. The factory is connected and uses the manufacturing data to improve automation, production monitoring and product quality [7]. BIIL is following the same principle in that it's collecting machine data and presenting the state of the factory on a central platform.

Schneider Electric – Smart Energy Monitoring in France. Schneider Electric have deployed Industrial IoT at their Le Vaudreuil factory in France. The WEF have reported that the factory used connected sensors and digital platforms to improve energy management and reduce material waste [8]. This is a very pertinent example of the kind of energy monitoring that BIIL's feature will provide (tracking of electricity use and the ability to identify waste).

Other Global Lighthouse Network cases. The World Economic Forum's The World Economic Forum's Global Lighthouse Network also has factories listed which are using AI for predictive maintenance (GlobalFoundries) [10], industrial IoT and smart manufacturing (Haier) [11], and AI-based quality and inventory improvement (Haier Qingdao) [12]. So again we see that BIIL's core functions of machine health monitoring, quality inspection, energy monitoring and tracking are all being used in industry.

2.2 OpenCV and Computer-Vision-Based Inspection

Product inspection is a classic application of Computer Vision in manufacturing and is at the heart of BIIL's design for garment and textile factories. This section looks at OpenCV and the research into automated fabric defect detection that is relevant to the proposed system.

2.2.1 Role of OpenCV in Industrial Vision

OpenCV is an open source computer vision library (Bradski [3]) with a wide range of functions including camera capture, filtering, thresholding, morphology, edge detection and contour analysis. It has a deep neural network module which can run trained detection models. It is a good fit for BIIL's edge device plan as it can run on low cost embedded computers (including Raspberry Pi class hardware with an AI accelerator) (Section 3.2).

2.2.2 Research on Fabric-Defect Detection

Guo et al. [5] have reviewed intelligent quality control of fabric surface defects and describe computer vision inspection as the key to intelligent manufacturing

in the textile industry. They categorise the detection methods into the traditional approaches (model, spectral, statistical and structural) and the learning based approaches (classical machine learning and deep learning) and go on to compare their advantages and disadvantages, datasets and real time performance.

Nasim et al. [6] looked at fabric defect detection under real manufacturing conditions (i.e. the images are of very low quality and not precisely positioned as you'd find on many benchmark datasets). Their YOLOv8n model achieved a mAP of 84.8% (precision 0.818, recall 0.839) across the seven defect classes. This demonstrates that deep learning can be used on actual production images and that the results on real factory data are of course significantly worse than on the clean benchmark datasets.

Zhao et al. [13] were focused on speed for online inspection. Their Dynamic Inference Network (DI-Net) allocated computation according to the complexity of the image and achieved 30.1 frames per second (20.96 m/min), some 20% faster than the base-line, with precision and recall both above 99% for normal samples and around 95% for defective samples. They point out that online detection won't be considered real-time until you're at the order of 15-20 m/min.

Carvalho et al. [4] developed an automated defect detection system for a weaving process in conjunction with a textile manufacturer. They went beyond just looking at model accuracy and linked model performance to operational metrics via Markov chain modelling. A sensitivity analysis was performed to determine the image acquisition frame rate which would result in the minimum number of false alarms.

Summary. the studies are predominantly using deep learning for detection. OpenCV can take care of the surrounding bits (image capture, pre-processing, region of interest selection, post-processing etc). BIIL combines the two.

2.2.3 Research Gap and Relevance to BIIL

The research gap identified from the reviewed applications and studies is:

1. Existing smart manufacturing systems are generally designed for the large factory environment and are not directly applicable to the cost, infrastructure and operational realities of the Bangladeshi RMG and textile sector.

2. Much of the existing work is focused on a single function (defect detection, predictive maintenance, energy monitoring etc) rather than integrating these functions into a single affordable platform.

3. Fabric defect datasets and models are often found to be unreliable when applied to real factory images due to the variation in lighting, fabric types, camera positions and production conditions.

4. Many of the proposed systems fail to integrate machine monitoring, quality inspection, production tracking, energy monitoring, workplace safety and factory analytics into one system.

5. The small and medium factory environment requires a modular solution which can start with a pilot line and then scale up without the need to replace their existing ERP or MES systems.

BIIL addresses these gaps by bringing together machine health monitoring, Opencv based defect detection, production tracking, energy monitoring, workplace safety and factory analytics into one platform for the manufacturing sector in Bangladesh. The aim is to create an affordable platform which can be used by both small and large factories.

2.2.4 Proposed Inspection Pipeline

The BIIL quality-inspection module is planned as a hybrid pipeline on the edge device:

1. **Capture:** an industrial network camera takes images of the fabric/product at a fixed point on the line.

2. **Preprocessing (OpenCV):** grayscale, noise filtering, illumination normalisation and region of interest selection.

3. **Rule-based checks (OpenCV):** thresholding, morphological operations and contour/edge analysis to flag up the obvious issues such as holes, stains, missing components and stitching breaks.

4. **Learned detection:** a lightweight object detection model (run via OpenCV DNN or an AI accelerator) to detect and classify more complex defects.

5. **Decision and reporting:** defect type, location and confidence are stored and sent to the dashboard with alerts to the Quality Inspector.

2.2.5 Comparison of Approaches

Table 2.1 compares the approaches considered for the inspection module. The comparison is based on the literature reviewed above.

Table 2.1: Comparison of defect-detection approaches considered for BIIL

Approach	Strengths	Limitations	Role in BIIL
Classical OpenCV pipeline (thresholding, morphology, edges, contours)	Fast and light on computation; needs no training data; easy to explain and tune on edge devices.	Sensitive to lighting and to fabric pattern or print changes; needs manual tuning per fabric; weak on subtle or irregular defects.	Capture, pre-processing, simple checks, fallback.
Deep-learning detector (e.g., YOLO family)	Handles varied defects and patterns; good results reported in recent studies [5, 6, 13].	Needs a labelled dataset; heavier computation; accuracy can drop on images unlike the training data [6].	Main defect locator and classifier.
Hybrid (OpenCV + Deep Learning)	Stable input from preprocessing; computation limited to relevant regions; simple rules stay explainable.	More components to maintain; needs tuning during the pilot.	BIIL quality-inspection module.

2.2.6 Key Challenges and Design Implications

- **Lighting and fabric variation.** Real factory images are very different from the benchmark images [6]. Controlled lighting, fixed camera position and OpenCV illumination normalisation are all part of the design.
- **Real-time speed.** The camera frame rate has to cover the moving fabric. For a fabric speed v and a camera field of view L in the direction of travel, full coverage requires at least v/L frames per second. As an example, at 20 m/min

($\approx 0.33\,\mathrm{m/s}$), which is within the real time range cited in [13], and with an assumed field of view of 0.25 m, you'd need at least about 1.3 fps; higher rates would allow for overlapping views and for repeated checks.

- **Limited labelled data.** Defect examples are both rare and varied [5]. We intend to collect and label images as part of the pilot and will make use of rule-based checks where we still have limited data.
- **False alarms and operational impact.** As [4] demonstrates, the frame rate and the metric chosen will have an effect on the number of false alarms and the production indicators. Alerts in BIIL will be accompanied by a confidence score so that the inspector can go on to assess them.
- **Edge hardware limits.** The model has to run on low cost edge hardware so we're favouring lightweight models over large ones.

2.2.7 Planned Evaluation

The module will be assessed in terms of precision, recall and mAP per defect class [6], processing speed in fps [13], and false alarm rate and its impact on production metrics [4].The results will be measured on the images collected from the pilot line during the AI-model and pilot phases of the timeline (Section 3.1).

Chapter 3

Plans and Deliverables

3.1 Timeline

The project covers 12 months and is divided into five phases. Table 3.1 gives the main output of each phase; Figures 3.1 and 3.2 show the project schedules.

Table 3.1: Project phases and main outputs

Phase	Months	Focus	Main outputs
1	1–2	Requirements & Design	Requirement specification, system architecture, use case and data flow diagrams, component selection.
2	3–5	IoT Deployment	Sensors, edge devices, and cameras installed; data acquisition and validation pipeline working.
3	6–8	AI Model Development	Anomaly detection and failure prediction models; OpenCV-based defect-inspection pipeline; model testing.
4	9–10	Dashboard & Pilot	Dashboard, reports, alerts, and ERP/MES link; pilot deployment and user testing.
5	11–12	Optimization & Commercialization	Performance optimization, model refinement, final documentation, and commercialization plan.

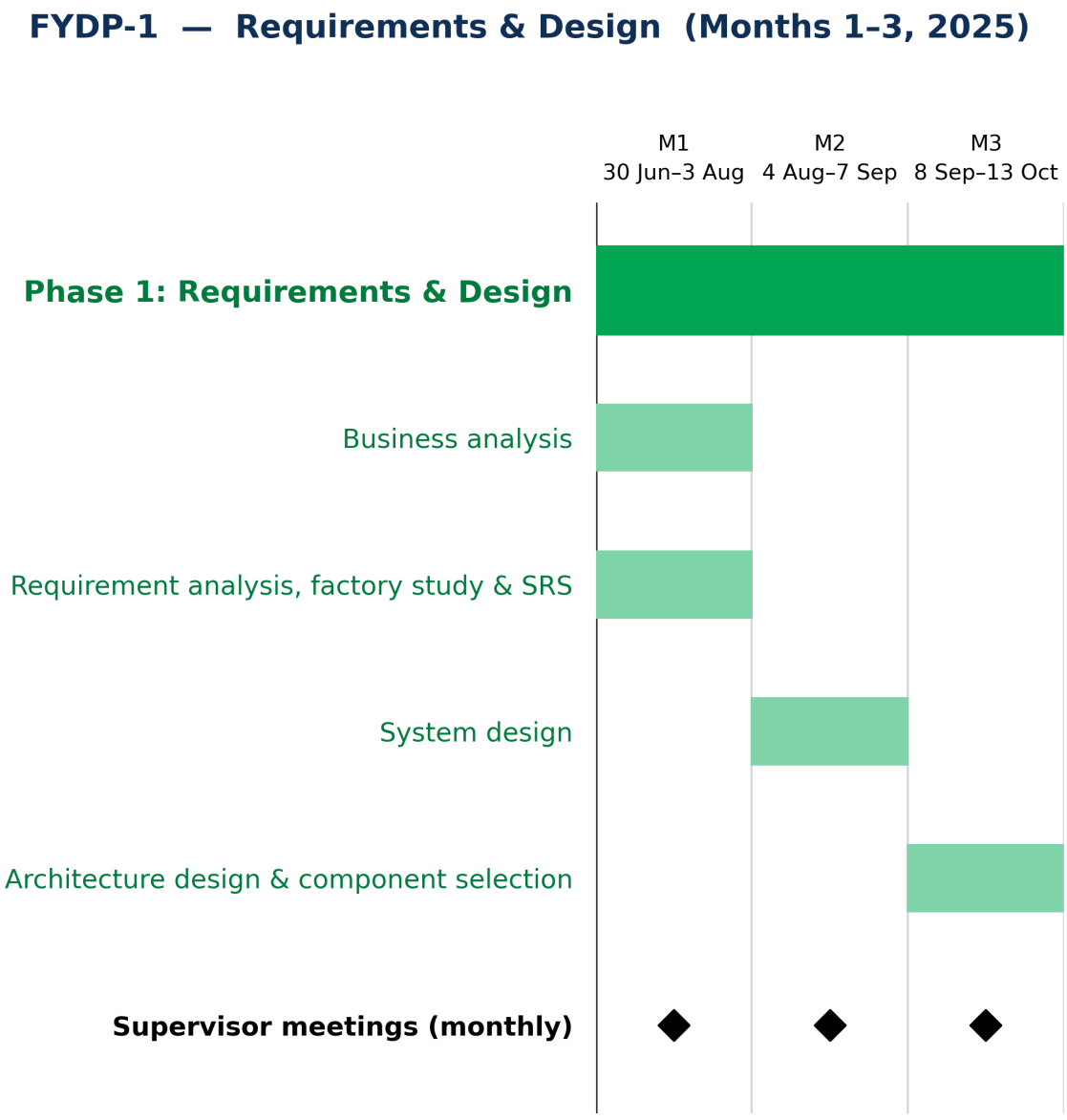


Figure 3.1: FYDP 1 Gantt chart

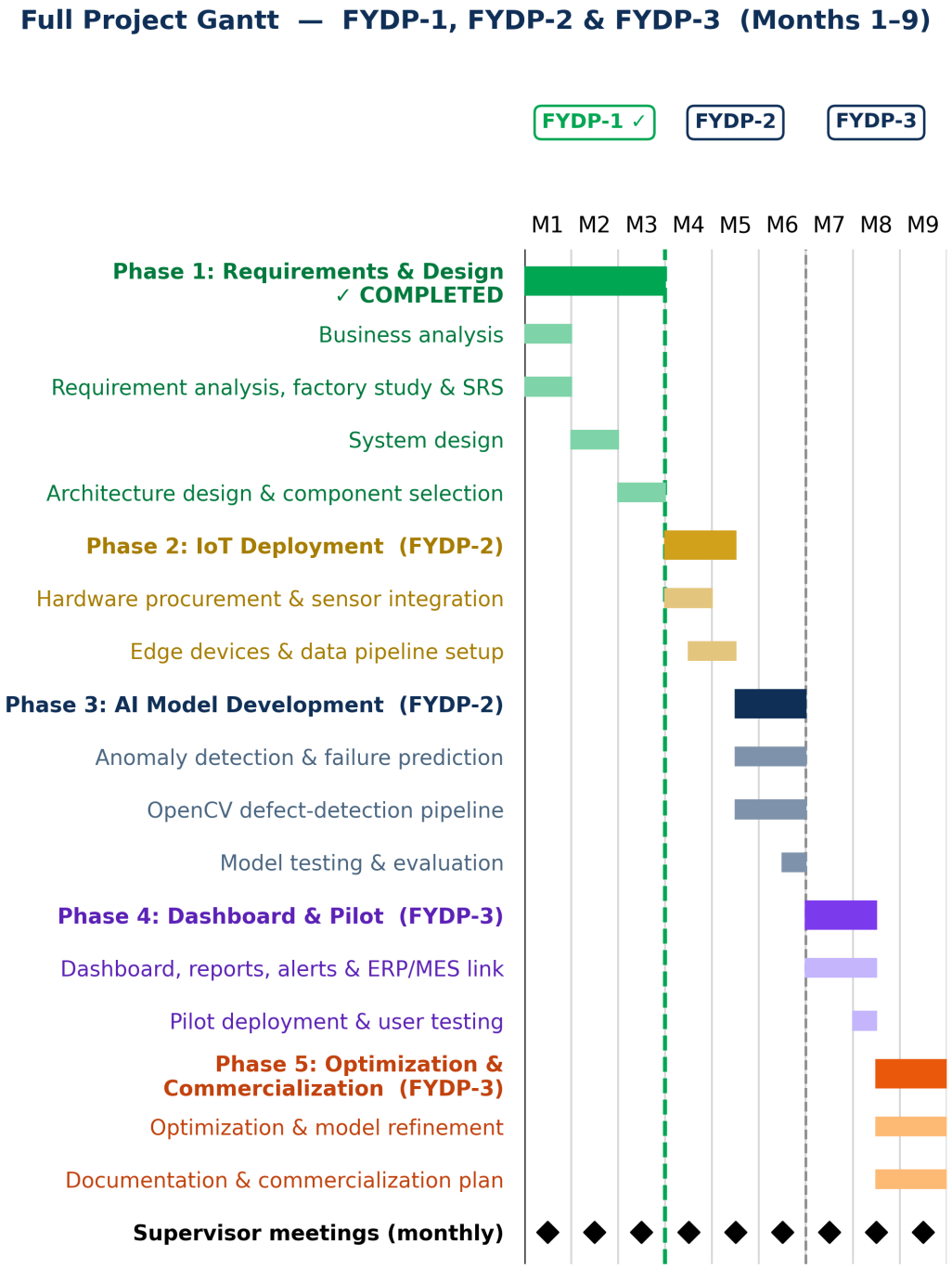


Figure 3.2: FYDP full 9-month Gantt chart

3.2 Budget

The estimated project cost is presented in two plans. The *Premium Budget* is the one with the larger team and the full hardware set. The *Standard Budget* is the one with the smaller team and the lower cost hardware. All figures are in BDT (bangladeshi Taka) unless otherwise stated. The item names in the tables are linked to the price sources.

Assumptions. Prices and exchange rate conversions are taken directly from the source pages provided and have not been independently verified. Import duties, shipping, taxes and subsequent price changes may well alter the final cost. Each subtotal is derived from the line items as listed. Enclosure, wiring and terminal costs are allowances given as ranges so the totals are also ranges. The figures are all based on the assumption that the team have nothing in the way of equipment or software already.

3.2.1 Budget Summary

Table 3.2 shows the two plans side by side . The Standard Budget is about BDT 921,830–922,830 (roughly 40%) less than the Premium Budget, primarily due to the smaller team and less expensive hardware .

Table 3.2: Summary of the main and alternative budgets (BDT)

Category	Premium Budget	Standard Budget
Human resources	1,980,000.00	1,245,000.00
Software and cloud	59,111.72	59,111.72
Hardware and equipment	276,705.00 – 278,705.00	88,875.00 – 91,875.00
Grand total	**2,315,816.72 – 2,317,816.72**	**1,392,986.72 – 1,395,986.72**

The main differences are:

- **Team:** the Standard Budget removes the separate Frontend Developer and Computer Vision Engineer roles (computer vision is covered by the AI/ML Engineer) and uses lower monthly rates.

- **Hardware:** the Standard Budget keeps the Raspberry Pi 5, AI HAT+, ESP32 controller, camera, temperature, gas, vibration, and energy sensing, and barcode scanner, but uses lower-cost models. It leaves out the Jetson Orin Nano, UPS, SSD, three-phase energy meter, PT100 temperature probe with converter, extra gas sensors, and isolated signal interface, so it covers a smaller monitoring scope suited to a pilot.

3.2.2 Premium Budget

Table 3.3: Premium Budget: human resources

Position	Qty.	Rate (BDT)	Duration	Cost (BDT)
Business Analyst	1	30,000/month	8 months	240,000.00
Frontend Developer	1	20,000/month	8 months	160,000.00
Full Stack Developer	1	30,000/month	8 months	240,000.00
AI/ML Engineer	1	50,000/month	8 months	400,000.00
Computer Vision Engineer	1	50,000/month	5 months	250,000.00
Embedded Systems Engineer	1	45,000/month	8 months	360,000.00
DevOps Engineer	1	45,000/month	4 months	180,000.00
QA / Test Automation Engineer	1	30,000/month	5 months	150,000.00
Human resources subtotal				**1,980,000.00**

Table 3.4: Premium Budget: software and cloud

Item	Description	Qty.	Rate	Duration	Cost (BDT)
Domain	.com registration	1	$10.98/year	1 year	1,365.16
Application Server	Cloud VPS Plus 8; 8 vCPU, AMD EPYC, 24 GB RAM, 450 GB NVMe, 1 Gbit/s port, unlimited traffic	1	$48/month	8 months	47,746.56
SSL Certificate	Let's Encrypt	1	Free	1 year	0.00
AI/ML API	AI advisory, recommendations, and model testing	Project allowance	—	Project allowance	10,000.00
Software and cloud subtotal					**59,111.72**

Table 3.5: Premium Budget: hardware and equipment

Item	Description	Qty.	Unit cost	Cost (BDT)
Raspberry Pi 5 8 GB	Single-board computer	1	35,600	35,600.00
Raspberry Pi AI HAT+ 2	AI accelerator HAT	1	25,000	25,000.00
UPS	Marsriva KP9 24,000 mAh mini UPS	1	11,000	11,000.00
SSD	MSI Spatium M470 Pro 500 GB SSD	1	14,500	14,500.00
Jetson Orin Nano Super 8 GB	Developer kit	1	99,500	99,500.00
Industrial ESP32-S3 RS-485 controller	Waveshare ESP32-S3 PoE ETH 8DI/8DO	1	5,360	5,360.00
Hikvision DS-2CD2T47G2-L	Network camera	1	18,350	18,350.00
Sensirion SHT85	Temperature and humidity sensor	1	4,560	4,560.00
MQ-2	Gas sensor module	1	160	160.00
MQ-5	Gas sensor module	1	170	170.00
MQ-135	Air-quality / gas sensor module	1	170	170.00
Isolated 4–20 mA / RS-485 interface	Industrial signal interface	1	5,000	5,000.00
SDM120 energy meter	Single-phase energy meter	1	4,200	4,200.00
SDM630 three-phase meter	Three-phase energy meter	1	25,545	25,545.00
MAX31865	SPI RTD PT100-to-digital converter module	1	550	550.00
PT100 probe	RTD temperature sensor, 100 mm probe	1	990	990.00
ADXL356 vibration module	3-axis vibration module (source link lists an ADXL346; verify exact model before purchase)	1	450	450.00

Item	Description	Qty.	Unit cost	Cost (BDT)
GM65 barcode module	QR / barcode scanner module	1	10,950	10,950.00
24 V DIN-rail PSU and protection	Mean Well TDR-240-24 power supply	1	9,650	9,650.00
Enclosure, wiring, terminals and other	Estimated allowance (range)	1	5,000–7,000	5,000–7,000
Hardware and equipment subtotal (including enclosure allowance)				**276,705.00 – 278,705.00**
Grand total (all listed line items)				**2,315,816.72 – 2,317,816.72**

3.2.3 Standard Budget

Table 3.6: Standard Budget: human resources

Position	Qty.	Rate (BDT)	Duration	Cost (BDT)
Business Analyst	1	20,000/month	8 months	160,000.00
Full Stack Developer	1	30,000/month	8 months	240,000.00
AI/ML Engineer (including CV)	1	40,000/month	8 months	320,000.00
Embedded Systems Engineer	1	35,000/month	8 months	280,000.00
DevOps Engineer	1	30,000/month	4 months	120,000.00
QA / Test Automation Engineer	1	25,000/month	5 months	125,000.00
Human resources subtotal				**1,245,000.00**

Table 3.7: Standard Budget: software and cloud

Item	Description	Qty.	Rate	Duration	Cost (BDT)
Domain	.com registration	1	$10.98/year	1 year	1,365.16
Application Server	Cloud VPS Plus 8; 8 vCPU Cores, AMD EPYC, 24 GB RAM, 450 GB NVMe, more storage available, 1 Gbit/s port, unlimited traffic	1	$48/month	8 months	47,746.56
SSL Certificate	Let's Encrypt	1	Free	1 year	0.00
AI/ML API	AI advisory, recommendations, and model testing	Project allowance	—	Project allowance	10,000.00
Software and cloud subtotal					**59,111.72**

Table 3.8: Standard Budget: hardware and equipment

Item	Description	Qty.	Unit cost	Cost (BDT)
Raspberry Pi 5 8 GB	Single-board computer	1	35,600	35,600.00
Raspberry Pi AI HAT+ 2	AI accelerator HAT	1	25,000	25,000.00
Industrial ESP32-S3 RS-485 controller	Waveshare ESP32-S3 PoE ETH 8DI/8DO	1	5,360	5,360.00
Dahua IPC-HFW2431SP-S	Network camera	1	6,500	6,500.00
SHT35 Temperature and Humidity Sensor 2pic	Temperature and humidity sensor module, I2C interface	1	1,250	1,250.00
MQ-135	Air-quality / gas sensor module	1	170	170.00

Item	Description	Qty.	Unit cost	Cost (BDT)
SDM120 energy meter	Single-phase energy meter	1	4,200	4,200.00
5V Fixed Output DC-DC Buck Converter	DC-DC buck step-down converter, constant voltage, 5V fixed output	1	170	170.00
801S Vibration Sensor	Vibration sensor, analog module	1	420	420.00
MH-ET LIVE Barcode Scanner V3.0	QR / barcode scanner module	1	3,250	3,250.00
24 V DIN-rail PSU and protection	DIN-rail switching power supply	1	4,955	4,955.00
Enclosure, wiring, terminals and other	Estimated allowance (range)	1	2,000–5,000	2,000–5,000
Hardware and equipment subtotal (including enclosure allowance)				**88,875.00 – 91,875.00**
Grand total (all listed line items)				**1,392,986.72 – 1,395,986.72**

3.2.4 Budget Notes

- **Premium Budget:** the vibration module listed as the ADXL356 has a source link which refers to an ADXL346 module so the exact module model needs to be confirmed prior to purchase.
- **Standard Budget, barcode scanner:** the MH-ET LIVE barcode scanner V3.0 is at BDT 3,250. The Sunlux XL-3700 2D handheld barcode scanner (BDT 3,400) is an alternative at BDT 150 more if you go down that route.
- **Standard Budget, temperature sensor:** the SHT35 is listed as ”2pic” at BDT 1,250 and is budgeted as a single line item. Need to confirm whether that

price is for two pieces.

3.3 Expected Final Product Description

The end product will be a working BIIL prototype for factory monitoring. It will comprise the edge hardware kit (with sensors, cameras, scanners and edge devices), the central BIIL platform and its AI models (including the OpenCV inspection module) and a web dashboard with reports and alerts. The intended users are factory managers, production supervisors, maintenance engineers, quality inspectors and factory floor staff in the garment, textile and other manufacturing sectors with a particular focus on small and medium sized factories. The results will be presented at the department's project exhibition and in this report.

3.3.1 System Overview

BIIL collects industrial data and turns it into information for factory operations. Sensors and cameras feed readings into the system which then checks, processes and stores the data. The data is used for machine monitoring, production tracking, quality inspection and safety monitoring. AI models will detect anomalies and provide predictions once sufficient data has been accumulated.

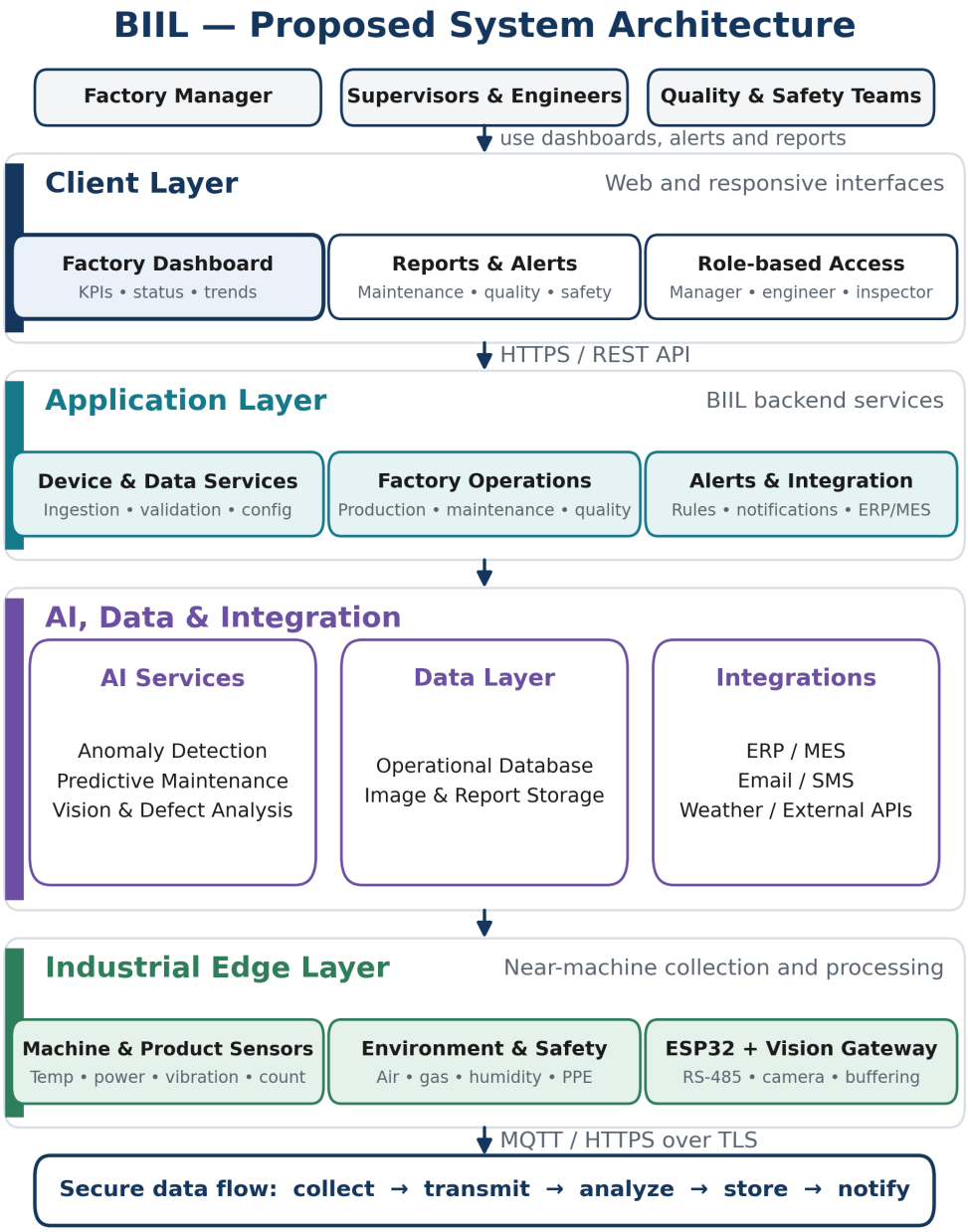


Figure 3.3: System architecture

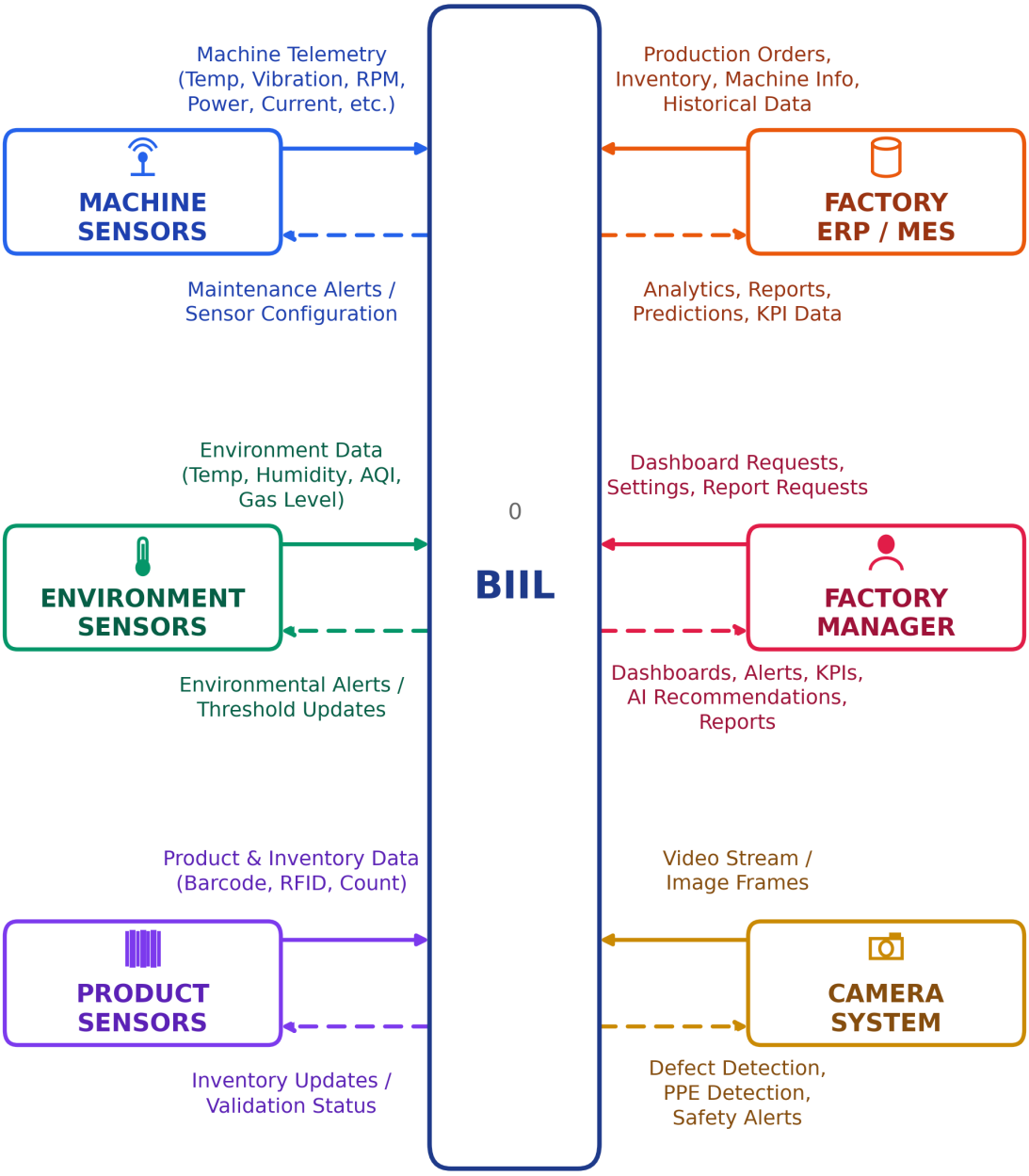


Figure 3.4: Data Flow Diagram – Level 0 (Context Diagram)

Description: The Level 0 DFD shows BIIL's interface with external devices, factory management and the ERP/MES systems. Machine sensors provide temperature, vibration, RPM and power. Environmental sensors provide temperature, humidity, air quality and gas level. Product sensors provide count and identity information and cameras provide images and video. BIIL takes all of this in and produces reports, alerts and other outputs.

3.3.2 Business Requirements

The business requirements are what BIIL is expected to provide to the factories. These are central monitoring, reduced machine downtime, improved production efficiency, better product quality and safer working environments. BIIL should also reduce energy waste, improve product traceability and provide reports to factory management. One of the requirements is that industrial digitalisation should be affordable and accessible in Bangladesh.

3.3.3 Functional Requirements

The functional requirements state what BIIL must do:

- **User Management:** Secure login and different access permissions for factory users.
- **Data Collection:** Take data from industrial sensors, cameras and product scanners.
- **Machine Monitoring:** Show machine status, detect abnormal conditions and support failure prediction.
- **Production Tracking:** Count the products and track them through production.
- **Quality Inspection:** Use Computer Vision (OpenCV-based) to identify product defects.
- **Energy and Safety:** Monitor energy usage, environmental conditions and PPE.

- **Dashboard and Reports:** Factory information, alerts and analytical reports. ERP/MES Integration: exchange of supported production and operational information with existing systems.
- **ERP/MES Integration:** Exchange of supported production and operational information with existing systems.

3.3.4 Non-Functional Requirements

The non-functional requirements specify the anticipated system qualities:

- **Performance:** The system must be able to process the industrial data at a rate that will allow for real time monitoring.
- **Scalability:** The system must be able to accommodate further machine, sensor and production lines as demand on the system grows.
- **Reliability:** The system must be able to carry on doing his core job and must be able to cope up with transient connection failure.
- **Security:** The system must secure the informations and enforce access control protocols.
- **Data Integrity:** The system must validate data that comes in to check that it doesn't create incorrect records or duplicate.
- **Usability:** The system must have dashboard that are clear and concise for user to understand.
- **Maintainability:** The system should be such that individual components can be updated and maintained in isolation.
- **AI Quality:** The system must use the approved AI model and continuously monitor their performance.

3.3.5 System Users

BIIL has five main user types, each with different factory responsibilities (Table 3.9).

Figure 3.5 shows their interactions with the system.

Table 3.9: BIIL system users and responsibilities

User	Responsibility
Factory Manager	Views factory performance, dashboards, and reports.
Production Supervisor	Monitors production lines and machine status.
Maintenance Engineer	Checks machine conditions, anomalies, and failure predictions.
Quality Inspector	Checks product quality and PPE compliance.
Factory Worker	Scans products using barcode or RFID systems.

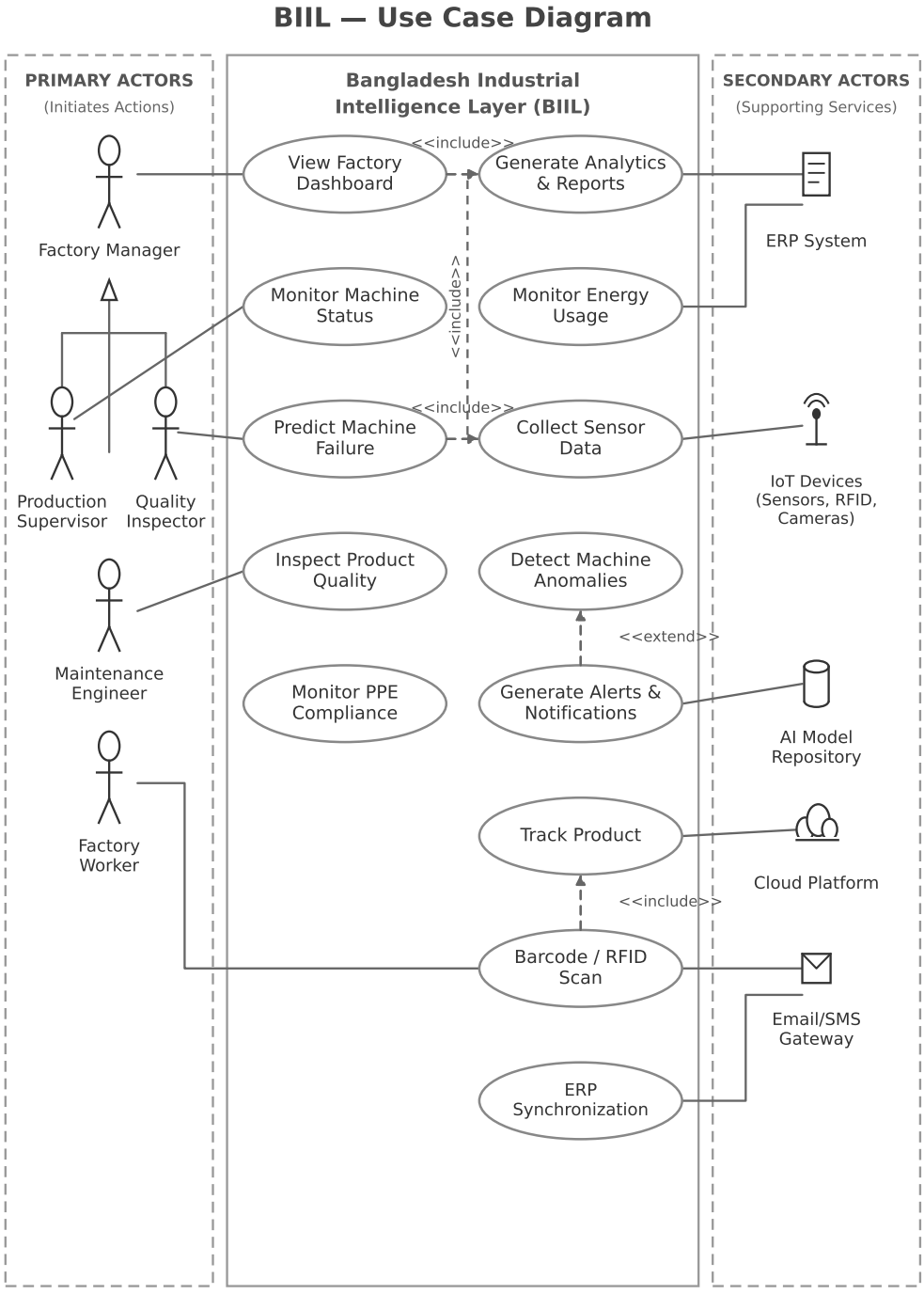


Figure 3.5: Use Case Diagram

Description: The use case diagram shows the various tasks assigned to the Factory Manager, Production Supervisor, Maintenance Engineer, Quality Inspector and Factory Worker. These are to view dashboards, monitor machines, check product quality, track products and receive alerts. There are also links to the IoT devices, the ERP system, the AI models and the notification service.

3.3.6 Technology Stack

BIIL is being designed as a modular software platform. The table below lists the various technologies that could be used at each of the layers. The technology that is ultimately selected will be the one that best meets the requirements of that component, the experience of the team, the integration needs, the cost of deployment and the level of maintainability. The frameworks listed are alternatives for various services and are not intended to be used in a single application.

Table 3.10: Proposed technology stack

Technology	Role	Alternatives
NestJS (Node.js backend)	Typescript based framework for modular APIs, auth, business logic and real time services.	Express.js, Fastify, or Spring Boot.
Django (Python backend)	Framework for data oriented services, admin interfaces, Rest APIs and integration with AI/ML workflows.	FastAPI, Flask, or NestJS.
MERN stack	Full stack JS/TSX with mongoDB, express.js, react and Node.js for the dashboard and app layer.	MEVN, PERN, or React with NestJS and PostgreSQL.
PHP (Laravel-based application)	Supports Webapp modules, dashboards, API endpoints and integration with existing PHP based systems.	Symfony, CodeIgniter, Django, or NestJS.

3.3.7 How the System Works

BIIL links together factory equipment, edge devices, central services and authorised users in a single information pipeline. Industrial sensors, energy meters, barcode readers and camera gathers machine condition reading, production counts, product identifiers, energy readings, environmental values, safety data and product/fabric images.

The data first gets to the edge layer. The edge controllers and edge computers receive the device signals, identify the source of the signal, check that the values are complete and in the correct format and do some initial processing. Camera images can be resized and filtered and fed into the OpenCV pipeline. Checking the data at this stage reduces the amount of network traffic and can flag up any communication or sensor issues prior to the data being sent to the central platform.

Validated records are passed through the factory network to the central BIIL platform where they are stored with timestamps, device IDS, production line details and all the other requisite metadata. The monitoring and AI functions then look for any unusual machine behaviour, any potential maintenance issues, any production changes, any visible defects, any energy waste and any safety events. BIIL can then pass on the relevant information to existing ERP and MES software so that it can support the current factory processes rather than replace them.

Dashboards, reports and notifications are the mechanism by which the results are delivered to the appropriate users. The factory manager can see the overall performance, the production supervisor can see the status of the line, the maintenance engineer can look at machine anomalies, the quality inspector can see the defect alerts and images. The workers and other users only see the information that is relevant to their role. The raw data, the processed data, the alerts and the decisions are all linked together through the monitoring process.

Chapter 4

Conclusion

The Bangladesh Industrial Intelligence Layer (BIIL) is being proposed as a more structured way of monitoring factory operations. The manufacturing sector is faced with all sorts of problems including machine breakdowns, product quality issues, energy waste and lack of production tracking. BIIL is intended to deal with all of these by collecting factory data and then analysing it using IoT, AI and Computer Vision.

By bringing all these activities onto one platform BIIL can help factory managers to see what's going on and react to it. The outcome would be a working prototype with real time machine and production monitoring, failure prediction, Opencv based defect detection, energy and safety monitoring and a dashboard with reports and alerts.

The key requirements, related work, initial system design, timeline and budget have all been put together. These will form the basis for development and testing in the coming stages.

References

1. BMW Group: This is how digital the BMW iFACTORY is: Sensor data and predictive maintenance. BMW Group Careers (2022), https://www.bmwgroup.com/en/news/general/2022/bmw-ifactory-digital.html

2. BMW Group: How AI is revolutionising production: AI-based quality inspection. BMW Group (2023), https://www.bmwgroup.com/en/news/general/2023/aiqx.html

3. Bradski, G.: The OpenCV Library. Dr. Dobb's Journal of Software Tools (2000)

4. Carvalho, A., Miguéis, V., Moura e Sá, M.: Enhancing operational performance in textile manufacturing: Impact of deep learning-based defect detection. Int. J. Prod. Res. 64(16), 6744–6768 (2026), https://doi.org/10.1080/00207543.2026.2632695

5. Guo, P., Liu, Y., Wu, Y., Gong, R.H., Li, Y.: Intelligent quality control of surface defects in fabrics: A comprehensive research progress. IEEE Access 12, 63777–63808 (2024), https://doi.org/10.1109/ACCESS.2024.3396053

6. Nasim, M., Mumtaz, R., Ahmad, M., Ali, A.: Fabric defect detection in real world manufacturing using deep learning. Information 15(8), 476 (2024), https://doi.org/10.3390/info15080476

7. Siemens: Digital manufacturing at Amberg: Electronics Works Amberg, Germany. Siemens (2023), https://www.siemens.com/en-us/company/insights/electronics-digital-enterprise-future-technologies/

8. World Economic Forum: Factories of the future and sustainability lighthouses: Schneider Electric smart factory. World Economic Forum (2022), https://www.weforum.org/media/factories-of-the-future-call-for-responsible-scaling-that-prioritizes-the-planet-and-people/

9. World Economic Forum: GlobalFoundries: AI-based predictive maintenance. World Economic Forum, Global Lighthouse Network (2025), https://www.weforum.org/media/global-lighthouse-network-2025-world-economic-forum-recognizes-12-new-sites-driving-holistic-transformation-in-manufacturing

10. World Economic Forum: Haier: Industrial IoT and smart manufacturing. World Economic Forum, Global Lighthouse Network (2025), https://www.weforum.org/

media/global-lighthouse-network-2025-world-economic-forum-recognizes-12-new-sites-driving-holistic-transformation-in-manufacturing

11. World Economic Forum: ACG packaging materials: Industrial IoT, AI and energy monitoring. World Economic Forum, Global Lighthouse Network (2026), https://initiatives.weforum.org/global-lighthouse-network/case-study-details/acg-packaging-materials---shirwal---productivity/aJYTG000000181x4AA

12. World Economic Forum: Haier Qingdao: AI-based quality and inventory improvement. World Economic Forum, Global Lighthouse Network (2026), https://initiatives.weforum.org/global-lighthouse-network/case-study-details/haier-strauss-science-%26-technology---qingdao---productivity/aJYTG00000018A14AI

13. Zhao, S., Zhong, R.Y., Xu, C., Wang, J., Zhang, J.: A dynamic inference network (DI-Net) for online fabric defect detection in smart manufacturing. J. Intell. Manuf. 36(4), 2881–2896 (2024), https://doi.org/10.1007/s10845-024-02387-2